

LATENCY-AWARE SELECTIVE MULTI-VIEW RETRIEVAL FOR UNSUPERVISED EVENT CLUSTERING

Kunqing Li^{1,2,3}, Lei Wang^{1,2,3,4,*}, Mengge Ai^{1,2,3}, Ruiteng Yan^{1,2,3},
Ke Wen^{1,2,3}, Enying Bao^{1,2,3}, Xiaoqiong Xu¹, and Xi Chen^{1,2,3,4,*}

¹College of Computer Science and Artificial Intelligence; ²Industrial College of Intelligent Services for IT Innovation

³The Key Laboratory for Computer Systems of State Ethnic Affairs Commission; ⁴School of National Security

Southwest Minzu University, Chengdu 610041, China; *Corresponding authors

{1053961104, 1521076843, 1427645033, 840380055}@qq.com

wangleiuestc@outlook.com, benying2001@163.com

xiaoqiongxu810@gmail.com, cx@swun.edu.cn

ABSTRACT

Streaming social-event monitoring requires bounded end-to-end latency, yet remote-LLM failures and retries create long critical paths. We propose MvSED, which minimizes time-to-memory-commit under a clustering-quality constraint rather than token count. Text fixes candidates, relevance-redundancy selects rerankers, and a frozen label-free gate repairs only uncertain assignments. Across two benchmarks, MvSED reduces wall-clock latency by up to 22.3%, preserves Event2012 clustering quality, and improves Event2018 ARI by 0.0567. Latency falls despite higher token counts, confirming that tokens are an unreliable proxy for end-to-end efficiency. Code: ghostelise.github.io/MvSED.

Index Terms— streaming text processing, latency optimization, retrieval-augmented reasoning, multi-view reranking, social event clustering

1. INTRODUCTION

Streaming social-media messages form asynchronous sequences of short textual observations. Real-time monitoring groups them into evolving events for crisis response and public-opinion analysis, where an accurate cluster delivered late may already describe a stale situation [1, 2]. Because event memory is updated sequentially after each block, delayed assignments postpone context for later blocks. End-to-end latency is therefore part of the monitoring objective, not merely an implementation statistic. Yet messages are sparse, informal, lexically diverse, and noisy; local graph structure alone may not associate separated reports of the same event [3, 4].

Retrieval-augmented generation (RAG) supplies complementary context from external memory [5]. RagSEDE applies it to unsupervised event detection and memory evolution [6], but prioritizes clustering quality rather than end-to-

end latency under a quality constraint. In an LLM-enhanced streaming pipeline, tokens measure payload, not critical-path time: embedding and retrieval, serial remote calls, service failures and backoff, and memory updates determine commit time. Fewer tokens therefore need not finish sooner; grounded multi-view evidence may spend more tokens yet avoid slow repeated interactions.

To address this latency-quality trade-off in streaming text processing, we propose MvSED, a latency-aware selective multi-view retrieval framework for unsupervised social event clustering. Rather than minimizing tokens, it minimizes end-to-end wall-clock latency subject to bounded quality degradation. Multi-view evidence becomes a *selective residual correction*: text fixes the admissible candidate set, a relevance-redundancy criterion selects auxiliary rerankers, and a deterministic label-free gate accepts repairs only for uncertain assignments satisfying support and margin constraints. Confident decisions bypass auxiliary reasoning; ambiguous anchors alone enter the bounded repair path.

Our central contribution is a critical-path-aware residual inference policy, not merely fewer calls. It controls which retrieval and reasoning operations enter the streaming pipeline, while fixed candidates and gated repair bound decision perturbations. MvSED thus optimizes measured time-to-memory-commit in Eq. (1) and treats tokens only as a diagnostic. With identical anchors, wall-clock latency falls by 11.3–22.3% even as token volume rises; Event2012 mean metric changes remain below 0.001, while Event2018 shows favorable observed changes. This exposes a systems-level distinction missed by token-only efficiency measures.

2. RELATED WORK

Social event detection. Early methods model lexical, user, or topic signals [7, 8]; later systems use graphs, incremental

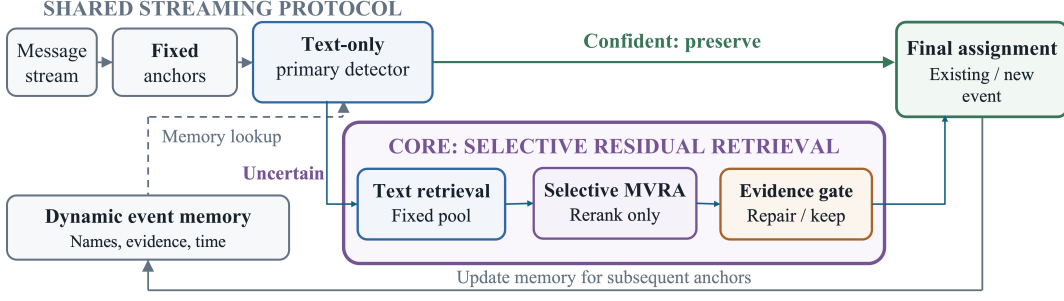


Fig. 1. Focused overview of MvSED. Confident assignments take the thin primary bypass; uncertain assignments enter the emphasized residual path of fixed-pool retrieval, selective MVRA, and evidence-gated repair.

learning, and structural entropy [3, 9, 4]. PromptSED uses label-assisted prompt learning and LSED a label-supervised classifier [10, 11], so their published scores are not mixed with our frozen label-free-inference comparison. We include HISEvent [4] as an external unsupervised quality reference. RagSEDE is the controlled baseline because MvSED retains its anchors, detector, LLM, and memory protocol, changing only residual retrieval and repair.

Retrieval and multi-view reasoning. Dense retrieval and RAG provide external evidence for downstream reasoning [12, 5]. Query rewriting can improve recall but may drift from the original intent [13]. Multi-view learning likewise requires explicit control of noisy and redundant views [14, 15]. MvSED combines these observations by separating candidate generation, auxiliary reranking, and repair acceptance.

3. METHOD

Let $\{m_j\}_{j=1}^N$ denote the messages in block t and $\mathcal{M}_{t-1}$ the event memory accumulated from previous blocks. The task assigns an open-set event identifier to every message. For policy π , the deployment objective is

$$\min_{\pi} T_{e2e}(\pi) \quad \text{s.t.} \quad Q(\pi) \geq Q_{\text{base}} - \epsilon, \quad (1)$$

where T_{e2e} is time-to-memory-commit and Q is clustering quality. This constrained streaming-system form minimizes response latency under a quality floor. Tokens are reported separately, not treated as a monotonic latency surrogate. Uncertain decisions activate fixed-pool MVRA and gated repair; confident decisions bypass them. Inference never accesses task labels, although the pretrained LLM contributes learned prior knowledge.

3.1. Protocol-aligned anchors and memory

We adopt RagSEDE’s single-stage anchors to keep inference units identical. Sentence-BERT embeddings [16] are grouped at fixed cosine threshold τ_a . Anchor capacity B_a , representative budget R , and representativeness–redundancy trade-off ρ

are selected during development and frozen; no second-stage reclustering is applied.

For each event, $\mathcal{M}_{t-1}$ stores its name, keywords, linked anchors, representative messages, and time observations. The text-only primary detector maps a_i to an existing event, OTHERS (novelty), or *Confusion* (abstention). Resolved events update memory.

3.2. Stage-aware selective multi-view retrieval

Multi-view reranking and alignment (MVRA) refines a text-grounded fixed pool. Representatives of a_i are concatenated and normalized into $q_i^{(t)}$, which retrieves the only admissible candidate set

$$\mathcal{C}_i = \text{TopK}_{K_c} \left(r_t(q_i^{(t)}, e), \mathcal{M}_{t-1} \right). \quad (2)$$

where r_t is raw SBERT cosine similarity. Auxiliary views $v \in \{\text{entity, relation, time}\}$ are generated from the same anchor and cannot introduce events outside $\mathcal{C}_i$. The fixed-pool constraint limits query drift to a reranking operation.

Let $\tilde{s}_v$ be view v ’s normalized score vector over $\mathcal{C}_i$. Text is always retained, and at most $L - 1$ auxiliary views are selected. For auxiliary view v , define its centered ranking-profile alignment A_v to text, redundancy R_v , score dispersion σ_v , and top-candidate consistency c_v by

$$A_v = \text{corr}(\tilde{s}_v, \tilde{s}_t), \quad R_v = \max_{u \in S_a} |\text{corr}(\tilde{s}_v, \tilde{s}_u)|, \quad (3)$$

$$g(v) = \sigma_v [c_v \max(0, A_v) - \lambda R_v].$$

Alignment A_v and consistency c_v suppress query drift, R_v penalizes overlapping evidence, and dispersion σ_v rewards discriminative rankings. Thus only text-consistent, useful, and nonredundant views enter the latency-critical path. A view is retained only if $A_v \geq \theta_{\text{align}}$ and $g(v) \geq \theta_{\text{gain}}$. Entity and relation views provide structural support. Time may resolve a ranking tie but cannot independently justify a repair. Candidate scores are fused with a text majority,

$$s_f(e) = \omega_t \tilde{s}_t(e) + (1 - \omega_t) \sum_{v \in S_a} \alpha_v \tilde{s}_v(e), \quad (4)$$

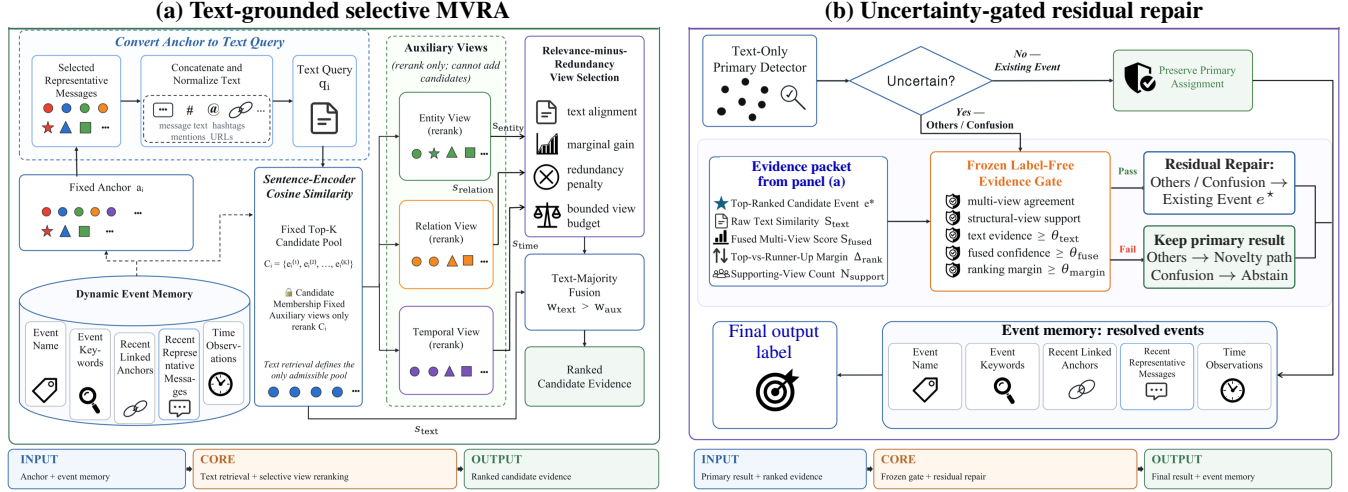


Fig. 2. Core method details: (a) text-grounded fixed-pool selective MVRA and (b) uncertainty-gated residual repair with memory maintenance. The shared inherited anchor protocol is described in Sec. 3.1.

where $\alpha_v = g(v) / \sum_{u \in S_a} g(u)$; if S_a is empty, $s_f = \tilde{s}_t$. The asymmetric weight $\omega_t > 1 - \omega_t$ preserves text as primary evidence.

3.3. Uncertainty-gated residual repair

The multi-view path is invoked only when the primary detector returns *OTHERS* or *Confusion*. Fig. 2(b) summarizes this residual path. Let e^* and e° denote the highest-scoring candidate and its runner-up, respectively. The gate accepts a reassignment to e^* if

$$G_i = [n_{\text{view}} \geq \eta_{\text{view}}] \wedge [n_{\text{struct}} \geq \eta_{\text{struct}}] \wedge [r_t(q_i^{(t)}, e^*) \geq \theta_{\text{text}}] \wedge [s_f(e^*) \geq \theta_{\text{fuse}}] \wedge [s_f(e^*) - s_f(e^\circ) \geq \theta_{\text{margin}}]. \quad (5)$$

View v supports e if e lies in its top- K_s over $\mathcal{C}_i$. n_{view} counts text/entity/relation support and n_{struct} entity/relation only; time is excluded. A passed gate assigns e^* without a second decision-stage LLM call. Rejection preserves the primary detector output, not the text-ranking Top-1: *OTHERS* follows the inherited novelty-creation path; *Confusion* remains an abstention label. Unresolved *Confusion* outputs share a reserved cluster ID within each block during evaluation. Selection/gating parameters are frozen after development; inference uses no task labels or extra training, and fixed K_c, L bound computation.

4. EXPERIMENTS

4.1. Experimental setup

Data and split. Event2012 contains 68,841 English tweets from 503 events [17]; Event2018 has 64,516 French tweets from 257 events [18]. We develop on Event2012 M1–M6,

test on M7–M21, and transfer selection/gate parameters unchanged to Event2018 M1–M16; only the pre-specified encoder and anchor threshold differ.

Models and controls. Event2012 and Event2018 use all-MiniLM-L6-v2 and multilingual DistilUSE, respectively, with one DeepSeek-R1-Distill-Qwen-32B endpoint on an RTX 4090. Within each panel, methods share blocks, encoder, endpoint, prompt, retry/decoding, seed, evaluator, and hardware; only the named module changes. Thus only RagSEDE supports paired latency and improvement claims; HISEvent is a quality reference. **Metrics and timing.** We macro-average NMI, AMI, and ARI [19] over blocks. The same script reconstructs message order and computes exact message-level cosine SC per block from saved embeddings and predicted clusters; block scores are macro-averaged. Because the frozen encoders differ, SC is compared only within each dataset, never across datasets. End-to-end wall-clock latency includes initialization, embedding/retrieval, remote inference, retry/backoff, and memory update; Proc. excludes initialization. Event2018 logs record aggregate LLM time, but not every remaining component share, which we do not infer. Calls and tokens are diagnostics. **Anchor verification.** Event2018 ordered hashes match. Event2012 rows match blockwise counts/configurations, but legacy files omit ordered hashes; this is not full-identity verification.

Frozen implementation. Anchor thresholds are 0.40 on Event2012 and 0.30 on Event2018; anchors contain at most 100 messages and three representatives. We set $K_c = 8$, $L = 3$, text/auxiliary weights 0.65/0.35, alignment/gain thresholds 0.05/0.015, and $\lambda = 0.35$. The gate requires two views, one structural view, text/fused scores 0.45/0.40, and margin 0.03. Generation allows 2048 tokens, 30 attempts, and 15-s retry delay. Source, frozen configurations, run wrappers, aggregations, and evaluators are public; datasets, weights, and raw

Table 1. Paired end-to-end efficiency and controlled ablations (R/M: RagSEDE/MvSED; bold: panel best; higher SC is better, and SC is compared only within each dataset).

Data	End-to-end latency			LLM calls		Failed calls		
	R/M (h)	Red.	Speed-up	R/M	Red.	R/M	Red.	
Event2012	11.15/9.89	11.3%	1.13×	2702/2668	1.3%	128/71	44.5%	
Event2018	23.22/18.04	22.3%	1.29×	4749/4468	5.9%	123/57	53.7%	
Combined	34.37/27.93	18.7%	1.23×	7451/7136	4.2%	251/128	49.0%	
<i>Controlled Event2012 ablation (M7–M21; equal anchor counts)</i>								
Method	NMI	AMI	ARI	SC	Proc. h	Calls	Failures	Lat. red.
RagSEDE	0.89593	0.88557	0.78158	0.30657	11.02	2702	128	–
MVRA	0.89470	0.88428	0.78443	0.29314	10.76	2741	51	2.41%
NoTime	0.89224	0.88167	0.77369	0.29812	11.65	2641	70	–5.73%
MvSED	0.89306	0.88199	0.77432	0.29859	9.78	2668	71	11.26%
<i>Aligned Event2018 ablation (M1–M16)</i>								
Method	NMI	AMI	ARI	SC	Proc. h	Calls	Failures	Lat. red.
RagSEDE	0.75418	0.74012	0.61094	0.09002	23.12	4749	123	–
MVRA	0.77229	0.76044	0.65022	0.09790	20.46	4753	80	11.49%
NoTime	0.75317	0.74126	0.63830	0.09005	17.06	4467	63	26.18%
MvSED	0.76400	0.75284	0.66762	0.09254	17.93	4468	57	22.45%

Table 2. Frozen-test clustering quality; HISEvent[†] is a non-matched external reference [4, 6].

Method	Event2012			Event2018		
	NMI	AMI	ARI	NMI	AMI	ARI
HISEvent [†]	0.80200	0.79200	0.61133	0.75375	0.74688	0.60375
RagSEDE	0.89637	0.88610	0.78386	0.75418	0.74012	0.61094
MvSED	0.89660	0.88615	0.78455	0.76400	0.75284	0.66762

logs are not redistributed.

4.2. Quality across evaluation scopes

Tables 1 and 2 share Event2012 blocks M7–M21 but report seed-42 and three-seed scopes, respectively. Three-seed NMI/AMI/ARI shifts are at most +0.00069; seed-42 decreases are at most 0.00726. Their apparent reversal therefore reflects seed aggregation, and both 95% intervals include zero. Event2018 gains are 0.0098, 0.0127, and 0.0567, with 10, 11, and 11 wins of 16; its intervals also include zero.

4.3. Latency and retry behavior

MvSED is faster on 14/15 Event2012 and 16/16 Event2018 blocks (bootstrap reduction intervals: 6.9–15.8% and 18.3–26.6%). Event2018 saves 5.18 h end to end and 5.26 h of LLM time with 66 fewer failures; without component timestamps, this supports only aggregate retry-path attribution.

RagSEDE/MvSED token totals (millions) are 1.789/2.077 on Event2012 and 3.602/4.030 on Event2018; Event2018 LLM time is 21.49/16.23 h (–5.26 h; –24.5%). Tokens rise 16.1%/11.9% while latency falls 11.3%/22.3%: token minimization would reject the earlier commit. Failure reductions of 44.5%/53.7% partly explain the reversal, but these aggregates are not a correlation estimate.

4.4. Controlled ablation

With matched anchors, **MVRA** changes Event2012 metrics by at most 0.0029 but raises Event2018 NMI/AMI/ARI by 0.0181/0.0203/0.0393. **NoTime** is fastest on Event2018 yet 5.73% slower on Event2012, so module count is not a wall-clock proxy; we do not attribute the inversion without component timestamps. Against NoTime, Event2018 Full spends 0.87 h but recovers 0.0108/0.0116/0.0293. **Full** cuts aggregate latency by 11.26%/22.45% and failures from 128/123 to 71/57; SC shifts are small (–0.0080/+0.0025) with intervals covering zero. On Event2018, 21.65% is the blockwise paired mean (CI 17.26–25.93%; 16/16 wins; $p = .000031$), whereas 22.45% is the aggregate-hours ratio. We claim consistent latency reduction, not universal quality gains.

5. CONCLUSION

MvSED optimizes time-to-memory-commit, not tokens. It cuts latency by 11.3% on Event2012 and 22.3% on Event2018 despite token increases of 16.1% and 11.9%. Event2012 quality changes stay below 0.001; Event2018 ARI rises by 0.0567, with intervals including zero. Broader tasks, endpoints, and seeds remain future work; we claim neither lower cost nor universal efficiency.

6. ACKNOWLEDGMENT

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7. COMPLIANCE WITH ETHICAL STANDARDS

We use the released Event2012 and Event2018 benchmarks. No new data were collected, and no human participants were recruited or subjected to intervention. The data were used in accordance with the terms of the original dataset providers; institutional ethics approval was not required.

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